When 10-K Risk Text Adds to Accounting Ratios: A Text-Tabular Dataset for Next-Year Deterioration

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Abstract

Annual reports contain both accounting ratios and a long risk-factor discussion. We ask whether that text carries information about next-year financial deterioration beyond the ratios, and whether adapting the text encoder to the label adds anything further. We join SEC financial statements to Item 1A of the EDGAR corpus, compress the risk section to ten specific lead sentences, and draw 1,000 firm-years. Five tabular learners are compared under structured, text-only, frozen-joint, and fine-tuned-joint conditions on five stratified folds. On the final table the mean joint gap is $+0.005$ AUROC and the mean fine-tuning gap is $+0.005$. Three of the five learners clear both gaps. A single fold had made the same recipe look several times larger, so the result is a small, split-sensitive gain rather than a large multimodal effect.

1 Introduction

A Form 10-K states the same year twice: once as financial statements and once as prose. Accounting ratios have long been used to separate failing firms from healthy ones (Altman, 1968). The risk-factor section, Item 1A, is where management describes what could make the next year worse. Those two views are not the same object. A ratio records the current balance sheet. A risk sentence records a contingency that may not yet be visible in the accounts.

Text-tabular learning asks whether the prose still helps once the numbers are available. Arazi et al. (2026) separate that question into two measurable gaps. The joint gap is the gain of a frozen text-and-table model over the better single modality. The awareness gap is the further gain from steering the text encoder toward the prediction target. A useful dataset should show both, and should not do so only because the text restates a column that is already in the table.

This project builds that dataset for next-year deterioration and evaluates it with the official MulTaBench protocol. The contribution is the curation, not a new model. We keep ratios that describe size, leverage, liquidity, and turnover, and we keep only the Item 1A sentences that are specific to the filing. Across five folds the two gaps are positive but small. The same table, read on the split used to design it, overstated both.

2 Related Work

Gradient-boosted trees remain strong tabular baselines. LightGBM grows trees leaf-wise (Ke et al., 2017). CatBoost reduces the bias of ordered boosting and handles heterogeneous columns (Prokhorenkova et al., 2018). TabM shows that a parameter-efficient ensemble of multilayer perceptrons can match more elaborate tabular networks (Gorishniy et al., 2025). TabPFN takes a different route: a transformer pretrained on synthetic tables performs in-context prediction on small datasets (Hollmann et al., 2025). TabPFN-2.5 extends that model to larger tables (Grinsztajn et al., 2025). MulTaBench evaluates exactly this mix, two tree models, TabM, and the two TabPFN versions, so that a dataset is not accepted because it suits only one learner (Arazi et al., 2026).

Unstructured inputs enter these models as embeddings. E5-small is a contrastively trained text encoder (Wang et al., 2022). A frozen encoder is optimized for general semantic similarity, so it can miss the wording that matters for one label. LoRA adapts a few low-rank weights without retraining the full network (Hu et al., 2022). MulTaBench applies that adaptation to the last three layers and calls the result a target-aware representation. The benchmark keeps a dataset only when, for at least three of five learners, the frozen joint model beats both unimodal models and the adapted encoder then beats the frozen joint model.

Financial text has its own line of work. Dictionary methods score the tone of a 10-K (Loughran and McDonald, 2011). Regression on the report text can predict market risk (Kogan et al., 2009). EDGAR-CORPUS supplies 10-K sections already split into items (Loukas et al., 2021). Those studies show that the filing contains predictive prose. They do not test whether that prose still helps after a small set of ratios is included, which is the comparison this dataset is built to support.

3 Method

3.1 Sources and join

Structured fields come from the SEC Financial Statement Data Sets, quarters 2015Q1 through 2021Q4, restricted to Form 10-K and 10-K/A with a full fiscal year. Text comes from EDGAR-CORPUS annual reports for 2016–2020 (Loukas et al., 2021). The tables are joined on the ten-digit CIK and the fiscal year. The 2021 statements supply the year-$t{+}1$ outcome for a 2020 filing, and 2015 supplies the prior year used to form changes. After the label filter below, 4,803 firm-years remain. We draw 1,000 of them uniformly with sample seed 42. The draw is between firms and years, not a firm-held-out split.

3.2 Label

For each firm-year we form three changes from year t to $t{+}1$: operating income, interest coverage, and operating cash flow. Each change is standardized on the pool, and the deterioration score is their mean. A lower score is worse. The worst 25% of the score is the positive class and the best 55% is the negative class. The middle 20% is discarded, so the label compares clear deterioration with clear health rather than neighboring percentiles. On the 1,000-row draw the

positive rate is 0.305. The label is a relative tail of this pool, not a bankruptcy filing.

3.3 Risk-factor text

Item 1A is used instead of the opening of Item 7. The opening of the management discussion is dominated by safe-harbor language, whereas Item 1A is the section that lists risks. A full Item 1A is far longer than the 512-token encoder window, and its first paragraphs are often a legend shared across filings. We therefore take the first sentence of each paragraph and apply four filters. A sentence must have 8 to 42 words and a digit fraction below 0.12. Sentences that contain a fixed list of boilerplate phrases are removed, including forward-looking statements, the securities-act legend, and generic cyber, personnel, and tax language. Sentences copied into more than 15% of filings are removed, as are sentences whose content words appear, on average, in more than 15% of filings. Of what remains, the first ten sentences in document order are kept. The CSV stores that excerpt without an extra encoder prefix. MulTaBench adds the E5 passage prefix itself.

3.4 Structured columns

The numeric columns are log total assets, debt to assets, the current ratio, cash to assets, and asset turnover. Margins, coverage, return on assets, growth, and capital expenditure were not used as inputs. Those quantities either restate the changes inside the label or tend to be narrated in the same words as the risk text. Capital expenditure to assets and fiscal year were removed after they failed to improve the average of the two gaps across folds. The current ratio and asset turnover are heavy-tailed, so each is replaced by $\operatorname{sign}(x)\log(1+|x|)$, rounded to three decimals. The other ratios stay on their winsorized scale, with the sample clipped at the 1st and 99th percentiles before that transform. Deleting rows, by contrast, was not helpful. Removing high-leverage firms or the smallest firms made the ratios easier and the text worse. The final table therefore changes columns, not the set of firm-years.

3.5 Evaluation

We use the MulTaBench evaluator without hyperparameter search. The five learners are LightGBM, CatBoost, TabM, TabPFN 2.0, and TabPFN-2.5. Each is trained in four conditions: ratios only, text only, both with a frozen E5-small encoder, and both after LoRA adaptation of the last three encoder layers. LoRA uses rank 16, learning rate 10^{-4}, at most 50 epochs, patience 3, and weight decay 10^{-3}. Embeddings are mean-pooled and reduced by PCA to 30 dimensions, fit on the training split only. Each fold holds out 10% of the rows with stratification. The fold index k uses random state $42+k$, so fold 0 is the split used during development. Condition means are rounded to three decimals before subtraction. For a learner, the joint gap is the frozen joint score minus the better unimodal score, and the awareness gap is the fine-tuned joint score minus the frozen joint score. The reported averages are the means of those five learner gaps.

Table 1: Five-fold mean AUROC on the final table. Gaps use means rounded to three decimals.

Model	Struct.	Text	Frozen	TAR	Δ_{J}	Δ_{A}
LightGBM	0.599	0.516	0.580	0.581	-0.019	0.001
CatBoost	0.601	0.541	0.608	0.596	0.007	-0.012
TabM	0.523	0.548	0.555	0.569	0.007	0.014
TabPFN 2.0	0.570	0.568	0.595	0.605	0.025	0.010
TabPFN 2.5	0.569	0.584	0.590	0.603	0.006	0.013
Average	0.572	0.551	0.586	0.591	0.005	0.005

4 Results

Table 1 is the five-fold mean on the final table. The average joint gap is $+0.005$ and the average awareness gap is $+0.005$. TabM, TabPFN 2.0, and TabPFN-2.5 have both gaps positive. LightGBM's awareness gap rounds to zero and its joint gap is negative, because the ratios alone score 0.599 against a frozen joint score of 0.580. CatBoost's frozen joint model beats its ratios by 0.007, and fine-tuning then gives that gain back.

The gaps are not stable across folds. Averaging the per-fold gaps, rather than subtracting the rounded five-fold means, gives joint gaps of $+0.021$, -0.021, $+0.020$, $+0.004$, and -0.033 on folds 0 through 4. Fold 0 is the development split. On that split an earlier version of the table, before fiscal year was removed and before the signed log, had average gaps of $+0.036$ and $+0.018$. The same earlier table, evaluated on all five folds, had average gaps of -0.001 and $+0.004$ (Appendix A). Text-only AUROC moves much more than the structured score. For

LightGBM on the final table, text-only ranges from 0.461 to 0.602 across folds, while the structured score stays between 0.558 and 0.652.

Two curation choices explain most of this pattern. First, the excerpt is short, about three sentences and 50 words at the median, because the frequency filter removes generic risk language. That text is informative on some splits and weak on others. When it is weak, several joint models follow it and fall behind the ratios, which turns the joint gap negative. Second, row filters that looked helpful on fold 0 removed the filings on which the text model had learned. High leverage and small firm size marked joint errors in one test split. Deleting those rows raised the structured score and lowered the text score on a refit. Changing the numeric columns, rather than the sample, was the edit that left both average gaps positive.

TabPFN 2.0 is the clearest success. Its frozen joint score of 0.595 exceeds both unimodal scores, and fine-tuning adds another 0.010. TabM is different. Its structured score is the weakest of the five, 0.523, so the joint gap is a recovery from a weak ratio model rather than a large lift over a strong one. Its awareness gap of 0.014 is real on the five-fold mean, but on individual folds the fine-tuned model often loses the frozen model's gain. With about 900 training rows, LoRA has little text to adapt and can memorize a split.

5 Conclusion

Item 1A, reduced to ten filing-specific leads, adds a small amount of information about next-year deterioration beyond size, leverage, liquidity, and turnover. Adapting E5 with LoRA adds a gain of the same size. Both statements are true for the five-fold mean of the final table, and both are carried by three of the five learners. They are not true for every fold. A result computed on the development fold is not a safe summary of this dataset.

The label is a percentile of three accounting changes, not a default or a delisting, and the 1,000 rows are a random draw rather than a split that holds out firms. The recipe was chosen using early folds of that draw, so the five-fold mean is not a completely untouched evaluation. Test folds contain about 100 rows, which is enough for the sign of a gap to flip. A next study should freeze the table before the reported folds, split by firm rather than by row, and test whether a longer excerpt can be added without returning the boilerplate that the frequency filter was built to remove.

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A Additional result tables

Table 2: Five-fold mean AUROC before removing fiscal year and before the signed log. Capital expenditure is already excluded. This is the table whose fold 0 looked strongest.

Model	Struct.	Text	Frozen	TAR	Δ_{J}	Δ_{A}
LightGBM	0.611	0.516	0.559	0.590	-0.052	0.031
CatBoost	0.628	0.541	0.611	0.594	-0.017	-0.017
TabM	0.542	0.548	0.587	0.572	0.039	-0.015
TabPFN 2.0	0.579	0.568	0.594	0.604	0.015	0.010
TabPFN 2.5	0.582	0.584	0.595	0.605	0.011	0.010
Average	0.588	0.551	0.589	0.593	-0.001	0.004

Table 3: Per-fold AUROC for LightGBM on the final table, illustrating the split sensitivity in Section 5. Scores are rounded to three decimals before the gaps.

Fold	Struct.	Text	Frozen	TAR	Δ_{J}	Δ_{A}
0	0.586	0.602	0.663	0.674	0.061	0.011
1	0.608	0.488	0.581	0.602	-0.027	0.021
2	0.558	0.461	0.518	0.555	-0.040	0.037
3	0.652	0.465	0.608	0.530	-0.044	-0.078
4	0.590	0.563	0.527	0.543	-0.063	0.016